# From ED Validation to a Real VQE Sweep

Seminar notes

August 10, 2026

## 1 Purpose of Week 7

Week 6 produced an ideal phase diagnostic: a non-local edge string swept across $\mu$ whose magnitude tracks the known topological interval $|\mu| < 2t$. That curve was trustworthy, but every plotted point was prepared by exact diagonalization (ED), not by a quantum circuit. The state on which the observable was measured was obtained by diagonalizing the $2^L \times 2^L$ Hamiltonian and reading off the lowest even-parity eigenvector.

Week 7 closes that gap. The milestone is:

> Replace the exact-diagonalization state preparation with a parity-constrained VQE preparation at every point of the $\mu$ sweep, and validate each prepared state against ED before any hardware noise is introduced.

Everything else from Week 6 is kept fixed: the same Hamiltonian, the same edge-string observable $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$, the same shot-measurement protocol, and the same $\mu$ grid. The only change is how the state is produced. This isolates a single question: can a variational quantum circuit reproduce the ideal diagnostic curve well enough to serve as the noiseless baseline for the Block 4 noise study?

## 2 What Changes Conceptually from Week 6

Week 6 answered "does the chosen observable reconstruct the phase diagram?" using a trusted ED reference state. Week 7 answers a different, harder question:

> "Can a gate-based, variationally optimized state reproduce that diagnostic across the whole parameter range?"

This is the step from a numerically exact target to an algorithm that must actually find the ground state. Three new difficulties appear that ED never had to confront:

1. the optimizer can fail or land in a local minimum;
2. the optimizer can drift into the wrong fermion-parity sector;
3. near criticality the ground state changes fastest, so a single ansatz must remain expressive enough everywhere.

Week 7 is largely about controlling these three failure modes and proving, point by point, that they did not occur.

## 3 Hamiltonian and Observable (Unchanged)

The Jordan–Wigner mapping of the open Kitaev chain gives

$$H(\mu) = -\frac{\mu}{2}\sum_{j=0}^{L-1} Z_j + \frac{t-\Delta}{2}\sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t+\Delta}{2}\sum_{j=0}^{L-2} Y_j Y_{j+1}. \tag{1}$$

As in Week 6 we use $L = 4$ and the sweet-spot couplings

$$t = \Delta = 1, \tag{2}$$

so the $XX$ term vanishes and

$$H(\mu) = -\frac{\mu}{2}\sum_{j} Z_j + \sum_{j=0}^{L-2} Y_j Y_{j+1}, \qquad \mu_c = \pm 2t. \tag{3}$$

The plotted quantity remains the magnitude of the parity-preserving edge string

$$|\langle O_{\mathrm{edge}}\rangle|, \qquad O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}, \tag{4}$$

which for $L = 4$ is $X_0 Z_1 Z_2 X_3$. The motivation for using a non-local string rather than a local edge operator is identical to Week 6: any operator anticommuting with parity has zero expectation value in a parity eigenstate, so only a parity-preserving, non-local observable is a stable diagnostic.

## 4 The VQE Sweep Workflow

The core of Week 7 is a loop over the $\mu$ grid. For each $\mu_k$ the workflow is:

1. build the Pauli-term Hamiltonian $H(\mu_k)$;
2. initialize the variational parameters from the previous point, $\theta_k \leftarrow \theta^*_{k-1}$ (warm start);
3. minimize the parity-constrained cost to obtain $\theta^*_k$;
4. validate the resulting state against the ED reference (energy, parity, subspace fidelity, edge-string error);
5. if validation fails, recover with random restarts and deeper optimization;
6. measure $|\langle O_{\mathrm{edge}}\rangle|$ both ideally and with finite shots;
7. store the parameters, diagnostics, and observable, then advance to $\mu_{k+1}$.

The single most important engineering change relative to Week 6 is that the optimizer is no longer restarted independently at each $\mu$. Neighboring ground states seed one another.

## 5 The EfficientSU2 Ansatz

VQE needs a parameterized quantum circuit because the ground state is no longer obtained by diagonalizing the Hamiltonian. Instead, we choose a family of trial states

$$|\psi(\theta)\rangle = U(\theta)|0\cdots 0\rangle, \tag{5}$$

and the classical optimizer changes the angles $\theta$ until the measured energy is minimized. The ansatz is therefore the search space of the algorithm: if it is too small, the correct ground state is not available no matter how good the optimizer is; if it is unnecessarily large, the optimization becomes slower and more unstable.

In the code this family is Qiskit's `EfficientSU2`, restricted to

```
EfficientSU2(L, su2_gates=['ry'], entanglement='linear', reps=reps)
```

This is a hardware-efficient ansatz: it is built directly from native-looking one-qubit rotations and nearest-neighbor entangling gates, rather than from a problem-specific exact diagonalization formula. In our restricted version the single-qubit layer contains only

$$R_Y(\theta) = \exp\left(-i\frac{\theta}{2}Y\right), \tag{6}$$

so each qubit gets a tunable real rotation. The entangling layer is a linear chain of CNOT gates, matching the one-dimensional geometry of the Kitaev chain. For $L = 4$ one entangling layer connects neighboring qubits as a nearest-neighbor ladder, schematically

$$(0,1),\ (1,2),\ (2,3), \tag{7}$$

up to Qiskit's drawing/endian convention.

This choice is useful for three reasons. First, the Kitaev Hamiltonian after Jordan–Wigner is a local nearest-neighbor qubit Hamiltonian, so a linear entangler is the natural circuit geometry. Second, the $R_Y$-only restriction keeps the state preparation real and greatly reduces the number of parameters compared with a full $R_X, R_Y, R_Z$ ansatz. That is appropriate here because the real matrix representation of the Hamiltonian admits real ED eigenvectors. Third, the CNOT layers are what allow the circuit to build the non-local correlations needed for the edge string $O_{\text{edge}}$; independent $R_Y$ rotations alone would only prepare a product state.

## 6 What `reps`=3 Means

The argument `reps` is the ansatz depth parameter. It does not mean that the whole VQE optimization is repeated three times. It means that the circuit pattern itself contains three repeated entangling blocks. In Qiskit's `EfficientSU2` layout, the structure is approximately

$$\left[R_Y \text{ layer}\right]\left(\left[\text{linear CNOT layer}\right]\left[R_Y \text{ layer}\right]\right)^{\texttt{reps}}. \tag{8}$$

Thus each repetition adds one more nearest-neighbor CNOT layer and one more layer of fresh $R_Y$ angles. For $L$ qubits with only $R_Y$ rotations, the number of variational angles is

$$N_\theta = L(\texttt{reps} + 1). \tag{9}$$

For the Week 7 sweep, $L = 4$ and `reps`$= 3$, so the optimizer controls

$$N_\theta = 4(3 + 1) = 16 \tag{10}$$

angles and the circuit contains three linear CNOT entangling layers.

The depth scan in plot 8 is precisely a test of this choice. With `reps`$= 1$, the circuit has only 8 angles and one entangling layer; it is often too rigid, especially near the critical point $|\mu| = 2t$. With `reps`$\geq 2$, the ansatz can already reach the correct low-energy state in this small $L = 4$ problem. We use `reps`$= 3$ in the full sweep as a conservative compromise: it gives one extra layer of expressivity beyond the minimum seen in the scan, while keeping the circuit shallow enough that the classical optimization remains cheap and the later noisy-circuit study is still meaningful.

# 7 Warm-Start Continuation

This section unpacks the three pieces of jargon used above: the star in $\theta_k^*$, the term "warm start," and the phrase "neighboring ground states seed one another."

## The variational circuit and the meaning of $\theta$

VQE does not store the quantum state directly. It stores a list of real numbers $\theta = (\theta_1, \ldots, \theta_p)$, the rotation angles of the gates in the ansatz circuit $U(\theta)$. Applying that circuit to the all-zeros state produces a trial state

$$|\psi(\theta)\rangle = U(\theta)|0 \cdots 0\rangle. \tag{11}$$

For the $L = 4$ `EfficientSU2` ansatz with `reps`$= 3$, each angle is the rotation of one $R_Y$ gate, and $\theta$ is the full vector of those angles. Choosing $\theta$ is choosing a state; optimizing $\theta$ is searching the state space the circuit can reach.

## Why $\theta_k$ carries a star

The bare symbol $\theta_k$ means "some parameter vector while we are still at point $\mu_k$," including the initial guess and all the intermediate guesses the optimizer tries. The starred symbol $\theta_k^*$ is reserved for the *converged, optimal* parameters: the specific angles the optimizer settles on after minimizing the cost at $\mu_k$,

$$\theta_k^* = \arg\min_{\theta} \; C_{\mu_k}(\theta), \qquad C_{\mu_k}(\theta) = \langle H(\mu_k)\rangle_\theta + \lambda\big(1 - \langle P\rangle_\theta\big)^2. \tag{12}$$

So $\theta_k$ is a moving guess during the search, and $\theta_k^*$ is the final answer at that point. The star is the standard "optimal value" notation, exactly as in $x^* = \arg\min f(x)$. The state $|\psi(\theta_k^*)\rangle$ is the VQE approximation to the ground state of $H(\mu_k)$, and it is the state on which the edge string is measured.

## What a "warm start" is

Any iterative optimizer needs a starting point. A *cold start* begins from a generic or random $\theta$ that carries no information about the answer; the optimizer must then travel a long way through parameter space to reach the minimum. A *warm start* begins from a point already believed to be close to the answer, so the optimizer only has to take a few small steps to converge.

In the $\mu$ sweep the warm start is the converged solution of the previous point:

$$\theta_k \xleftarrow{\text{initialize}} \theta_{k-1}^*. \tag{13}$$

Concretely, after we finish optimizing at $\mu_{k-1}$ and obtain $\theta_{k-1}^*$, we do not throw those angles away. We hand them to the optimizer at $\mu_k$ as its initial guess and let it refine them. Because the optimizer starts near the answer, it needs far fewer iterations and is much less likely to wander off into a wrong local minimum.

### Why "neighboring ground states seed one another"

The justification is physical, not just numerical. The ground state $|\psi(\mu)\rangle$ of the Kitaev chain changes *smoothly* with $\mu$ everywhere except at the critical points $|\mu| = 2t$. If two sweep points $\mu_{k-1}$ and $\mu_k$ are close, their ground states are nearly the same state, so the angles that prepare one are nearly the angles that prepare the other:

$$\mu_k \approx \mu_{k-1} \implies |\psi(\mu_k)\rangle \approx |\psi(\mu_{k-1})\rangle \implies \theta_k^* \approx \theta_{k-1}^*. \tag{14}$$

"Seeding" is just the act of passing $\theta_{k-1}^*$ forward as the seed (initial guess) for the next optimization. "Neighboring ground states seed one another" therefore means: because adjacent ground states are physically close, each converged solution is an excellent initial guess for its neighbor, and the sweep can be run as one continuous chain

$$\theta_0^* \to \theta_1^* \to \theta_2^* \to \cdots \tag{15}$$

rather than as a set of unrelated optimizations. This is the technique called *continuation* (or homotopy): solve an easy point first, then track the solution by taking small steps in the parameter $\mu$. The same warm-start chain can also be run in both directions across the grid, so that each point benefits from a well-converged neighbor.

### Cold vs. warm in the sweep

**Independent random (cold) starts — the Week 5/6 prototype style.**

- repeat the full, expensive optimization from scratch at every point;
- can converge to different local minima at adjacent $\mu$, producing artificial jumps;
- make the swept curve noisy even before shot noise is added.

**Warm-start continuation in $\mu$.**

- initialize $\theta_k$ from $\theta_{k-1}^*$;
- track the smooth evolution of the ground state with only a few optimizer steps per point;
- keep random restarts only as a fallback when validation fails.

### What "deeper optimization" means

When a warm-started point fails validation, the recovery path does "deeper optimization." This does *not* mean a deeper circuit; it means spending more optimization effort to dig out of a bad local minimum. In practice it combines:

- **more iterations** — raise the optimizer iteration cap (`maxiter`) so it is allowed to keep refining instead of stopping early;
- **multiple restarts** — launch several independent random initial guesses and keep the best converged result, instead of trusting the single warm-started run;

- **a tighter parity penalty / re-selection** — re-run while enforcing the even sector so the optimizer cannot escape into the wrong-parity minimum.

Only if even this fails would one consider a genuinely deeper ansatz (more `reps`); the depth scan in the next-to-last section shows `reps`$= 3$ is already sufficient away from a handful of hard points, so "deeper optimization" is almost always about effort, not circuit depth.

### Where it is hardest

The expected difficulty is concentrated near $|\mu| = 2t$, where the bulk gap closes, the ground state changes fastest, and the smoothness assumption behind warm starts is weakest. There the previous solution $\theta^*_{k-1}$ is a poorer seed, the optimizer is most likely to stall, and the recovery path (restarts plus deeper optimization) is most likely to trigger. This is exactly where the fidelity checks earn their place.

## 8 Parity Must Be Controlled

The Hamiltonian conserves fermion parity,

$$P = \prod_{j=0}^{L-1} Z_j, \qquad [H, P] = 0. \tag{16}$$

In the topological phase the even- and odd-parity sectors are nearly degenerate. Plain energy minimization can therefore return different sectors at different $\mu$, and a sector flip looks like a spurious discontinuity in $\langle O_{\text{edge}} \rangle$.

The Week 7 strategy is to bias the optimization toward a fixed (even) sector with a parity penalty:

$$C(\theta) = \langle H \rangle_\theta + \lambda \big(1 - \langle P \rangle_\theta\big)^2. \tag{17}$$

The penalty is minimized when $\langle P \rangle = +1$, pushing the optimizer into the even sector without distorting the energy landscape inside that sector. The parity expectation $\langle P \rangle$ is recorded at every point, and states that fall outside the parity tolerance are rejected and recovered.

## 9 Validation Dashboard

A low VQE energy alone is not a sufficient acceptance criterion near degeneracy. Each swept point must pass four independent checks against the ED reference:

| **Metric** | **Question** | **Purpose** |
|---|---|---|
| Energy error | Is $E_{\text{VQE}}$ near $E_{\text{ED}}$? | Detect optimizer failure |
| Parity | Is $\langle P \rangle \approx +1$? | Keep one sector across the sweep |
| Subspace fidelity | Does the VQE state overlap the correct low-energy space? | Handle near-degeneracy correctly |
| String error | Does $\langle O_{\text{edge}} \rangle$ match the ED value? | Validate the quantity actually plotted |

For a non-degenerate ground state, ordinary fidelity is

$$\mathcal{F} = |\langle \psi_{\text{ED}} | \psi_{\text{VQE}} \rangle|^2. \tag{18}$$

This asks whether the prepared VQE state is the same vector as the ED target, up to an irrelevant global phase. In the topological region, however, the low-energy physics is nearly degenerate: the even- and odd-parity ground states can become exponentially close in energy. Then overlap with one arbitrary eigenvector can be misleading, because ED may choose a particular basis inside a nearly degenerate space while the variational circuit may produce another equally valid vector in the same physical space.

The more robust diagnostic is subspace fidelity. If $\{|\phi_a\rangle\}$ spans the accepted ED low-energy subspace, define the projector

$$\Pi_{\text{sub}} = \sum_{a \in \text{sub}} |\phi_a\rangle\langle\phi_a|, \tag{19}$$

and measure

$$\mathcal{F}_{\text{sub}} = \langle\psi_{\text{VQE}}|\Pi_{\text{sub}}|\psi_{\text{VQE}}\rangle = \sum_{a \in \text{sub}} |\langle\phi_a|\psi_{\text{VQE}}\rangle|^2. \tag{20}$$

This number is close to one when the VQE state lies in the correct low-energy manifold, even if it is not aligned with one particular ED eigenvector.

For the Week 7 sweep we additionally pin the even-parity sector with the penalty term. Therefore the implemented validation compares against the ED even-parity ground state; in this fixed-sector setting, the accepted subspace is effectively the one-dimensional even ground space. The reason we still track and plot it as a fidelity/subspace diagnostic is that it protects us from the main failure mode near degeneracy: a low energy alone could be correct while the state has leaked into the wrong sector or into the wrong combination of low-energy states. A point that fails any check triggers deeper ansatz repetitions, more optimizer iterations, or random-restart recovery.

## 10 Implementation Shape

The loop body, in schematic form, is:

```
theta = initial_parameters()
for mu in mu_values:
    H = qiskit_hamiltonian(L, t, mu, delta)
    theta = optimize(parity_cost(H), initial=theta)

    diagnostics = validate(theta, H, ed_reference(mu))
    if not diagnostics.pass_all:
        theta = recover_with_restarts(H)

    string_value = measure_edge_string(theta, shots=4096)
    save(mu, theta, diagnostics, string_value)
```

The implementation lives in `src/block3_core.py` (helpers `vqe_ansatz`, `vqe_cost`, `evaluate_state`, `solve_point`, `vqe_sweep`, `measure_edge_string_shots`, `best_state`, `depth_scan`) and is driven by `src/block3.py --plots 7` (the sweep) and `--plots 8` (the depth diagnostic). The default command-line setting is `--reps 3`, which creates the 16-parameter, three-entangler ansatz described above. A practical point: the sweep must persist optimizer parameters and per-point diagnostics, not only the final observable, so that failed regions are debuggable and reusable as warm starts for later noise studies.

# 11 Reading the Week 7 Sweep Figure

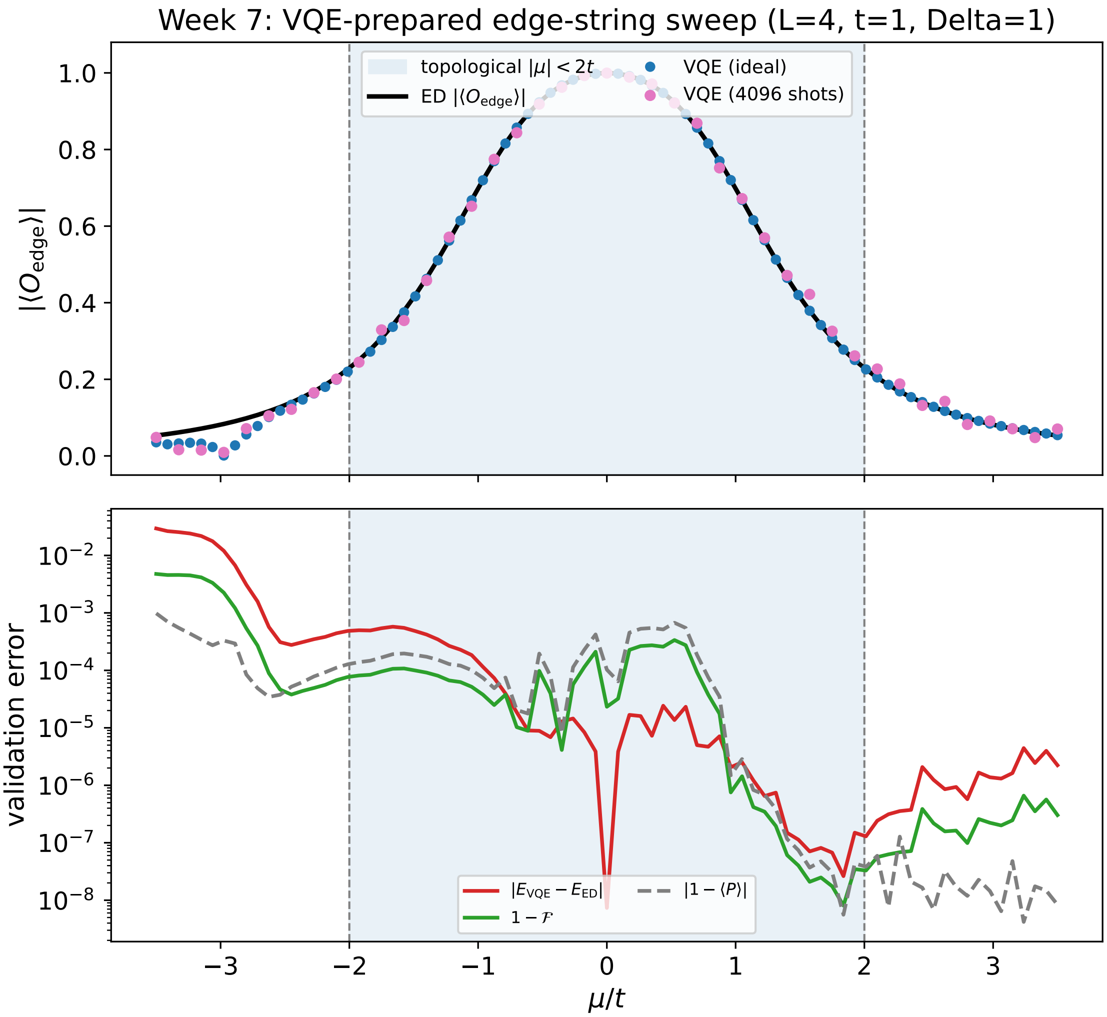


Figure 1: Week 7 VQE-prepared sweep for $L = 4$, $t = \Delta = 1$. Top: the edge string $|\langle O_{\text{edge}} \rangle|$ from parity-constrained VQE, both ideal and shot-based, plotted against the ED baseline. Bottom: per-point validation showing energy error, infidelity $1 - \mathcal{F}$, and parity deviation $|1 - \langle P \rangle|$; crosses mark points that required random-restart recovery.

The expected behavior is:

- the VQE curve (ideal and shot-based) overlays the ED baseline across the whole $\mu$ grid;
- energy error and infidelity stay near zero inside the topological region and rise mildly near $|\mu| = 2t$;
- parity deviation stays small everywhere because the penalty pins the even sector;
- recovery markers, if any, cluster near criticality where the optimization is hardest.

The top panel is the measurement-side result that Block 4 will compare against; the bottom panel is the certificate that the result was obtained from genuinely correct state preparation.

## Ideal VQE, Shot-Based VQE, and the Validation Curves

The label *VQE ideal* means that the optimized VQE circuit is converted to its noiseless statevector and the edge-string expectation value is evaluated directly:

$$\langle O_{\text{edge}}\rangle_{\text{ideal}} = \langle\psi(\theta^*)|O_{\text{edge}}|\psi(\theta^*)\rangle. \tag{21}$$

There is no sampling error in this curve. It answers: if the variational circuit prepared exactly the state described by its final parameters, what edge-string value would it give?

The label *VQE 4096 shots* uses the same optimized circuit and the same observable, but estimates the expectation value from 4096 simulated projective measurements. For an observable with eigenvalues $\pm1$, the statistical scale is at most

$$\sigma \sim \frac{1}{\sqrt{4096}} \simeq 0.016, \tag{22}$$

with smaller variance when the true value is already close to $\pm1$. Thus this curve should fluctuate around the ideal VQE curve; it is not a different state-preparation method.

For the Week 7 edge string,

$$O_{\text{edge}} = X_0Z_1Z_2X_3, \tag{23}$$

the shot method is implemented as a real measurement protocol. First the optimized VQE circuit prepares

$$|\psi(\theta^*)\rangle = U(\theta^*)|0000\rangle. \tag{24}$$

Then the qubits appearing with $X$ in the observable are rotated into the hardware $Z$-measurement basis. Since

$$X = HZH, \tag{25}$$

Hadamard gates are applied to qubits 0 and 3 before measurement. After these basis rotations, measuring

$$X_0Z_1Z_2X_3 \tag{26}$$

is equivalent to measuring the product

$$Z_0Z_1Z_2Z_3 \tag{27}$$

on the rotated circuit.

Each measured bitstring therefore gives one sample $o_m = \pm1$. Write the measured bits as $b_j \in \{0,1\}$. For $L = 4$,

$$o_m = (-1)^{b_0+b_1+b_2+b_3}. \tag{28}$$

An even number of measured 1 bits gives $+1$, while an odd number gives $-1$. The shot-based estimator is the sample average

$$\langle O_{\text{edge}}\rangle_{\text{shots}} = \frac{1}{N_{\text{shots}}}\sum_{m=1}^{N_{\text{shots}}} o_m, \qquad N_{\text{shots}} = 4096. \tag{29}$$

This is the same estimator that would be used on real hardware: prepare the optimized VQE state, rotate the measurement basis, measure many times, convert bitstrings to $\pm1$, and average.

The lower panel is not another phase diagnostic. It is a validation panel. In the current plotting convention the red curve is the energy error

$$|E_{\text{VQE}} - E_{\text{ED}}|, \tag{30}$$

while the other curves show the state-fidelity error $1 - \mathcal{F}$ and the parity deviation $|1 - \langle P \rangle|$. A small red value near $\mu = 0$ means that the ansatz and optimizer found the ED even-sector ground energy accurately there. This is expected because the sweet-spot/topological state is easy for the $R_Y$ plus linear-CNOT ansatz to represent.

If the red validation curve is larger on the negative-$\mu$ side and decreases as the sweep proceeds toward positive $\mu$, that asymmetry should be read as an optimization-history effect, not as new physics. The physical ED Hamiltonian and the edge-string diagnostic are symmetric under the corresponding $\mu \leftrightarrow -\mu$ transformation at this sweet spot. The VQE sweep, however, is ordered: it begins at the left edge of the grid from a random initial parameter vector and then uses warm-start continuation, where each optimized point seeds the next one. Early points can therefore carry larger optimization error, while later points benefit from a better parameter path. In short, the top-panel ED physics should be symmetric, but the bottom-panel validation errors need not be exactly symmetric because they measure how the classical optimizer performed along one directed sweep.

## 12 Ansatz Depth: How Deep Is Enough?

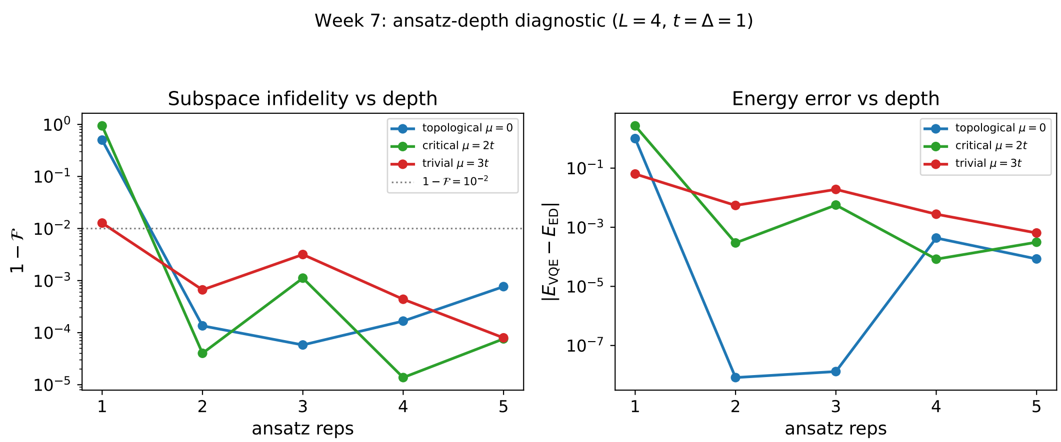


Figure 2: Ansatz-depth diagnostic: multi-start VQE subspace infidelity and energy error versus the number of ansatz repetitions, at three representative $\mu$ (topological, critical, trivial).

Scanning the number of `EfficientSU2` repetitions establishes the minimum depth needed for faithful preparation:

- one repetition (8 parameters) is too shallow; it fails worst at criticality $\mu = 2t$ (fidelity $\sim 0.06$);
- two or more repetitions already reach the even-parity ground space;
- the critical point is consistently the hardest, exactly where the bulk gap closes.

This justifies the choice `reps`$= 3$ used in the full sweep: deep enough to clear criticality with margin, shallow enough to keep the optimization cheap.

## 13 Definition of Done for Week 7

- a scripted $L = 4$, $t = \Delta = 1$ VQE sweep over the same $\mu$ grid as Week 6;

- consistent even-parity preparation across the entire sweep;
- ED comparison for energy, parity, subspace fidelity, and edge-string expectation at every point;
- a shot-based VQE edge-string curve plotted against the ED baseline;
- optimization cost and failure/recovery statistics recorded per $\mu$ point.

## 14 Bridge to Block 4

Week 7 validates quantum state preparation; it does not yet test robustness. Only after the ideal VQE sweep passes validation should noise be introduced. The Block 4 plan is:

- freeze the validated ideal VQE sweep as the noiseless reference;
- apply realistic gate, readout, and relaxation ($T_1$, $T_2$) noise to the preparation and measurement circuits;
- separate optimizer degradation from measurement degradation;
- compare raw and error-mitigated edge-string curves;
- quantify the noise level at which the inferred phase boundary becomes unreliable.

The conceptual progression across the block is now complete in outline: Week 6 validated the diagnostic with ED, Week 7 validates the quantum state preparation, and Block 4 tests whether the full circuit workflow survives realistic noise.